

Password Security Audit Tool Report

1. Objective

The objective of this project is to develop a Password Security Audit Tool using Python. The tool evaluates the strength of a password based on various security parameters such as length, uppercase letters, lowercase letters, numbers, and special characters. It helps users create stronger and more secure passwords.

2. Tool Used

- Python Programming Language
- Visual Studio Code (VS Code)
- Windows Operating System

3. Methodology

The Password Security Audit Tool was developed using Python. The program accepts a password as input from the user and evaluates its strength using predefined criteria.

The following parameters are checked:

- Password length should be at least 8 characters.
- Presence of uppercase letters.
- Presence of lowercase letters.
- Presence of numeric digits.
- Presence of special characters.

Based on these checks, the password is categorized as:

- Weak Password
- Medium Password
- Strong Password

4. Source Code

```
password = input("Enter Password: ")

score = 0

if len(password) >= 8:
    score += 1

if any(char.isupper() for char in password):
    score += 1

if any(char.islower() for char in password):
    score += 1
```

```
if any(char.isdigit() for char in password):
    score += 1

special_chars = "!@#$%^&*()_+=[{}|;:'.<>?/"

if any(char in special_chars for char in password):
    score += 1

if score <= 2:
    print("Weak Password")
elif score <= 4:
    print("Medium Password")
else:
    print("Strong Password")
```

5. Test Results

Test Case 1

Input Password:
abc123

Output:
Weak Password

Test Case 2

Input Password:
Tanushree123

Output:
Medium Password

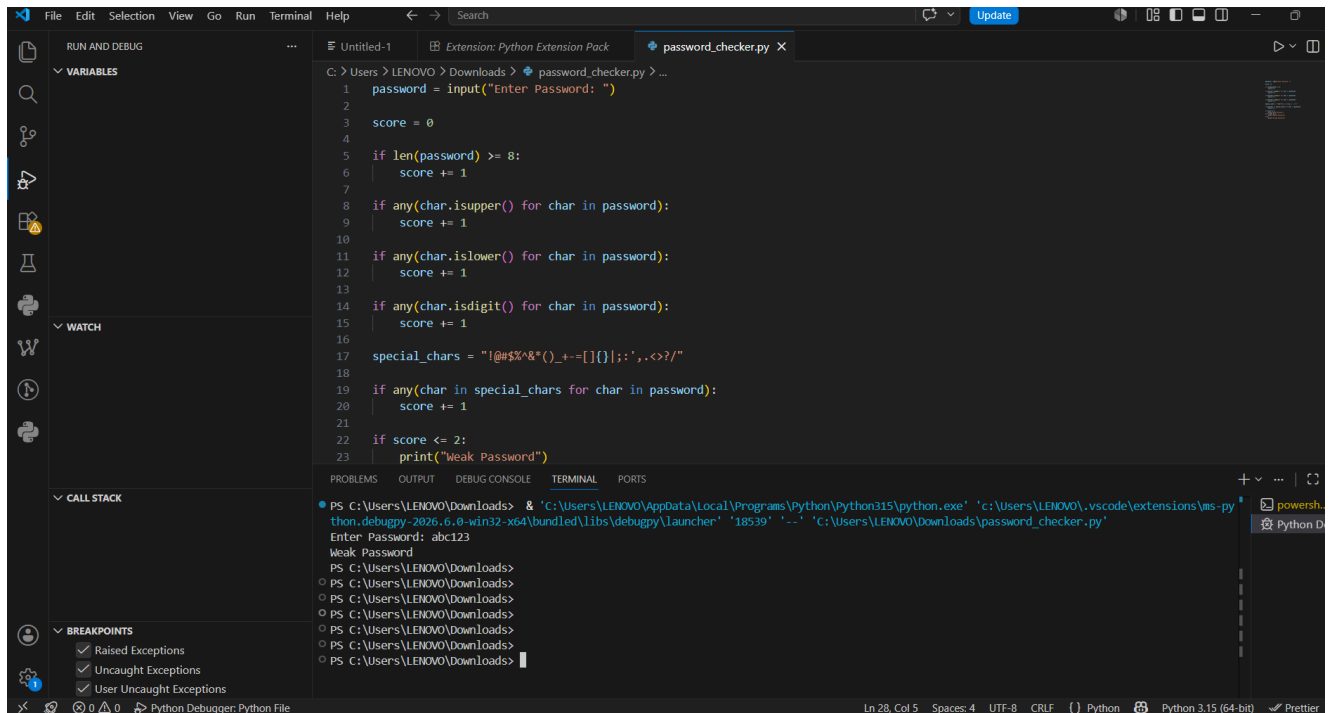
Test Case 3

Input Password:
Tanushree@123

Output:
Strong Password

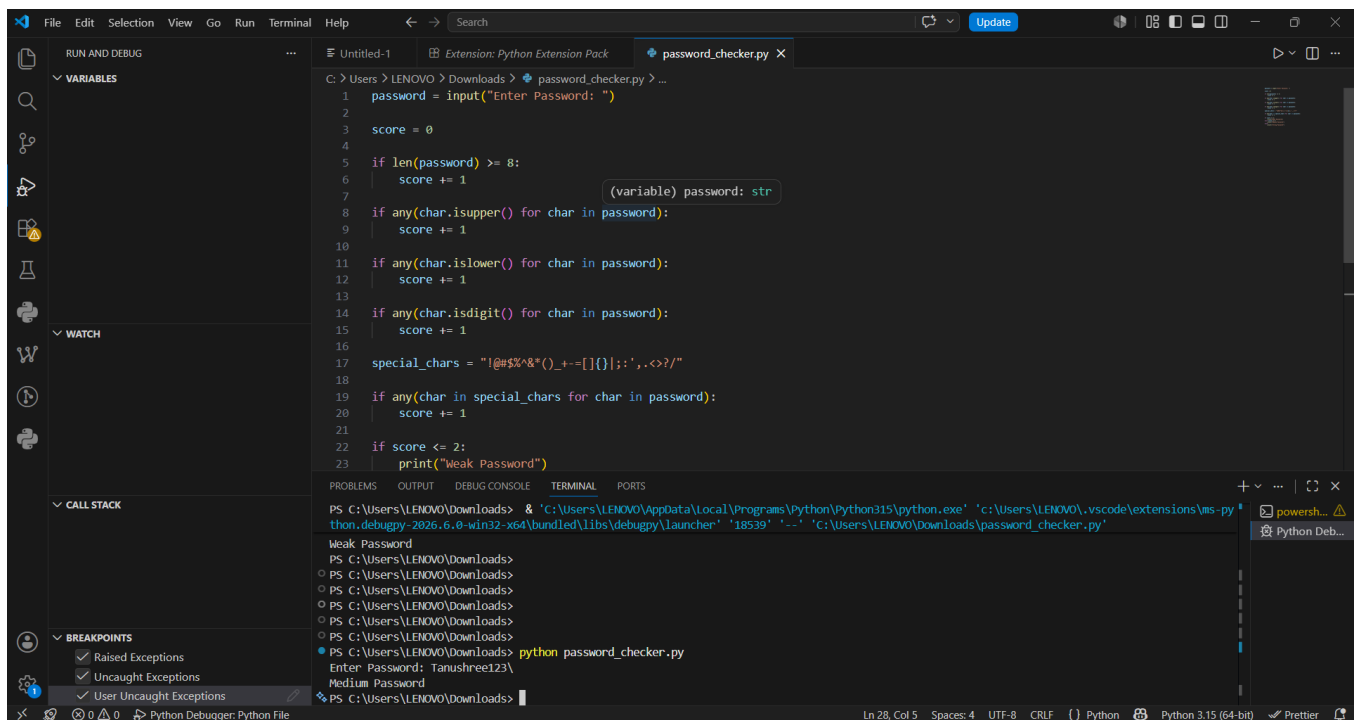
6. Screenshots

Screenshot 1



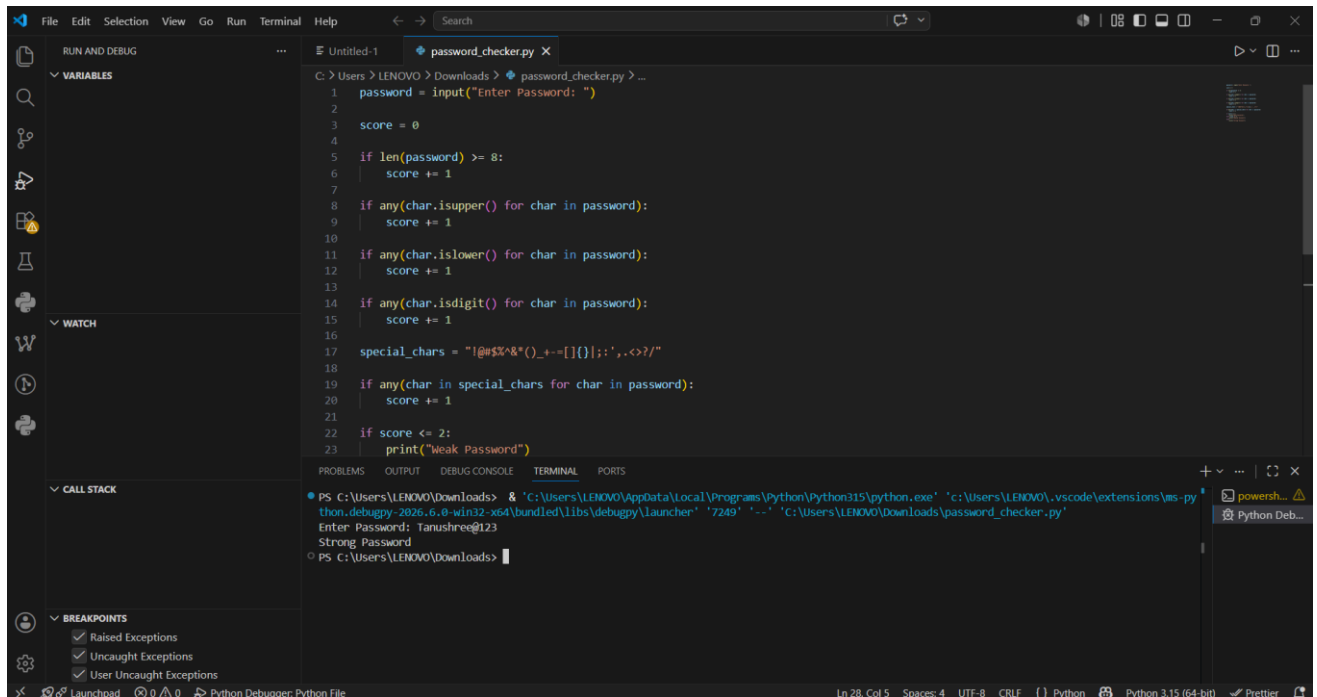
Weak Password Result

Screenshot 2



Medium Password Result

Screenshot 3

A screenshot of the Visual Studio Code editor with a Python file named 'password_checker.py' open. The script is a password strength checker. It prompts the user to enter a password, then evaluates it based on several criteria: length (must be at least 8 characters), presence of uppercase letters, lowercase letters, digits, and special characters. A score is calculated based on these factors. If the score is 2 or less, it prints 'Weak Password'. Otherwise, it prints 'Strong Password'. The terminal at the bottom shows the command prompt running the script, and the output shows 'Enter Password: Tanushree@123' followed by 'Strong Password'. The left sidebar shows the 'RUN AND DEBUG' panel with sections for VARIABLES, WATCH, CALL STACK, and BREAKPOINTS. The bottom status bar indicates the file is a Python 3.15 (64-bit) file using Prettier for formatting.

```
File Edit Selection View Go Run Terminal Help
password_checker.py X
C:\Users\LENOVO>Downloads>password_checker.py>...
1 password = input("Enter Password: ")
2
3 score = 0
4
5 if len(password) >= 8:
6     score += 1
7
8 if any(char.isupper() for char in password):
9     score += 1
10
11 if any(char.islower() for char in password):
12     score += 1
13
14 if any(char.isdigit() for char in password):
15     score += 1
16
17 special_chars = "!@#$%^&*()_+~[]{}|;:',.<>?/"
18
19 if any(char in special_chars for char in password):
20     score += 1
21
22 if score <= 2:
23     print("Weak Password")
24
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\LENOVO\Downloads> & 'C:\Users\LENOVO\AppData\Local\Programs\Python\Python315\python.exe' 'c:\Users\LENOVO\.vscode\extensions\ms-python.debugpy-2026.6.8-win32-x64\bundle\libs\debugpy\launcher' '7249' '-' 'c:\Users\LENOVO\Downloads\password_checker.py'
Enter Password: Tanushree@123
Strong Password
PS C:\Users\LENOVO\Downloads>
Ln 28, Col 5 Spaces: 4 UTF-8 CRLF Python Python 3.15 (64-bit) Prettier
```

Strong Password Result

7. Conclusion

The Password Security Audit Tool was successfully developed using Python. The tool effectively evaluates password strength based on multiple security factors and categorizes passwords into Weak, Medium, and Strong levels. This project improved understanding of password security concepts, input validation, and Python programming fundamentals. Strong passwords play an important role in protecting user accounts and sensitive information from unauthorized access.